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EDUCATION

University of North Carolina at Charlotte Masters of Computer Science	GPA : 3.7 Aug 2024 – Dec 2025
SRM University, AP Bachelors of Technology, Computer Science and Engineering	GPA : 3.24 Jun 2019 – May 2023

EXPERIENCE

Graduate Assistant- Full Stack Developer | University of North Carolina at Charlotte | Charlotte, NC Aug 2024 - Dec 2025

- Designed a healthcare management platform featuring appointment scheduling, EHR management, automated notifications, and analytics dashboards, increasing patient interaction tracking by 35%.
- Developed 10+ RESTful APIs using Spring Boot with WebSocket support for real-time updates, optimized PostgreSQL using Hibernate, and built Angular-based web interfaces serving 500+ users.
- Implemented secure authentication and role-based access control using JWT, OAuth 2.0, and Keycloak, significantly reducing unauthorized access across campus systems.
- Developed responsive and accessible UI components using React, Tailwind CSS, and Material-UI, improving module accessibility and driving a 25% increase in daily student usage.
- Configured CI/CD pipelines with Jenkins and orchestrated Dockerized microservices deployment using Kubernetes on AWS (EC2, S3, RDS, Lambda), supporting concurrent users with high availability and minimal downtime.

Software Developer | HealthEdge Aug 2023 – Jul 2024

- Addressed 20+ production-critical defects in healthcare applications built with Java, Spring Boot, and SQL, reducing customer-reported incidents by 30% and improving SLA adherence.
- Streamlined SQL queries, PL/SQL procedures, and stored functions within claims and billing modules, achieving a 45% reduction in query execution time and improving financial transaction accuracy.
- Refactored legacy codebases using SOLID principles and Design Patterns, reducing cognitive complexity from over 100 to under 12 and improving long-term maintainability.
- Created automated testing pipelines using JUnit, Mockito, PowerMock, TestNG, and GitLab CI/CD, increasing unit and integration test coverage from 60% to 82%.
- Enhanced observability using log4j, ELK Stack, and AWS CloudWatch to identify and resolve system issues, strengthening monitoring and debugging across application workflows.

Software Developer Intern | HealthEdge Jan 2023 – Jul 2023

- Engineered backend services using Java (Spring Boot) and Python (Flask), integrating RESTful APIs for EHR and claims workflows, and executing SQL scripts to support healthcare data processing.
- Generated unit and integration tests using JUnit, PyTest, Postman, and SOAP UI, reducing defect leakage by 28% and improving overall module reliability.
- Optimized SQL queries, stored procedures, and database schemas in MySQL and PostgreSQL, achieving a 38% improvement in average execution time for high-volume financial and healthcare modules.
- Containerized microservices using Docker and deployed them through GitLab CI/CD with Shell scripting, actively contributing within Agile teams and applying TDD practices during sprint cycles.

Research Intern | SRM University Jan 2023 – May 2023

- Designed a modular multi-sensor pedestrian tracking system using accelerometer, gyroscope, and magnetometer data, applying supervised learning and time-series signal processing to reduce sensor noise and drift, resulting in a 37% improvement in indoor navigation accuracy.
- Conducted feature extraction, error analysis, and model tuning for dead reckoning algorithms, reducing misclassification rates by 33% and improving location consistency across 15+ indoor environments to support augmented reality and location-based services.

PROJECTS

Healthcare Management System | Java, Spring Boot, Hibernate, MySQL, RESTful APIs, HTML, CSS, JavaScript

- Created a full-stack healthcare system with patient records, appointments, and billing workflows by implementing secure Spring Boot RESTful APIs with Hibernate, a responsive front end, and robust validation, improving operational efficiency by 35%.

Localization Quality Estimation Model | PyTorch, Transformers, Linguistic Feature Extraction, Cloud Dashboard

- Devised a translation quality estimation model using transformer embeddings and linguistic feature extraction, implementing a confidence-scoring pipeline that reduced manual QA review efforts by 50%.

Statistical Analysis CLI Tool Enhancement in Java with JUnit Testing | UNIX/Linux, Java, Junit, Eclipse, Git

- Revamped a Java-based CLI statistical analysis tool by expanding functionality to include standard deviation and user-driven inputs, operated in UNIX/Linux environments and validated with JUnit using Eclipse IDE.

TECHNICAL SKILLS

C#,Python, Java, C, C++, R, Bash, PHP, TypeScript, JavaScript, SQL (Postgres, MySQL, MS-SQL), HTML, CSS, XML, JSON, Spring Boot, Node.js, Angular, React.js, Django, Bootstrap, jQuery, JUnit, Mockito, TestNG, PyTorch, TensorFlow, AWS (EC2, S3, RDS, Lambda, IAM, API Gateway, ELB, CloudFormation), Docker, Kubernetes Jenkins, GitLab CI/CD,ELK Stack, Prometheus, MongoDB, DynamoDB, Cassandra, RESTful APIs, OAuth 2.0, gRPC, Microservices Architecture, Cloud Computing, Distributed Systems, Operating Systems, Database Normalization, Machine Learning, API Design, Software Testing, Agile (Scrum), Maven, Postman, JIRA, Git, Eclipse, IntelliJ IDEA.